SIH26264 v2: Upgrade Plan

Integrated Dashboard for Monitoring and Tracking Institution-Related Complaint Cases Across AICTE Bureaus/Cells. Team Glorious Purpose. Prepared 28 Oct 2026.

1. Straight assessment first

- **What I could not check:** the Vercel app rendered as an empty page to my fetcher (client-side app), the GitHub repo shows only a *frontend* folder with 3 commits (no backend visible), and I cannot open the Drive video. So I have NOT seen your UI. Everything below about v1 weaknesses is inferred from the repo layout and your own description, not observed.
- **Likely v1 gap (inferred):** a frontend-only Vercel deploy cannot back claims like real-time counts, audit history, RBAC or overlap detection. If v1 runs on mock data, judges who probe will find it. v2's job is to make every claim true and demonstrable.
- **Biggest risk is time, not ideas.** A full rebuild from scratch in a few days will fail if scope is unrestrained. The plan below has a hard cut line (Section 7). Build the PS core first, add-ons second, AI last.
- **What gets you shortlisted** is rarely feature count. It is: every PS requirement visibly demonstrated, honest synthetic-data labelling, an overlap detector with measured results, and a workflow an officer understands in one demo. Features that do not serve those get cut.

2. What the research verified (and what it did not)

Finding	Source / status	Why it matters for the build
AICTE PGRC handles: student/faculty/stakeholder grievances, CPGRAMS grievances, ragging complaints, norm-violation complaints, VIP/Ministry references, court matters, Parliament questions, RTI.	aicte.gov.in PGRC page (retrieved). Verified.	Real case sources and deadline types. Justifies source_channel and hard-deadline tracking. Court/PQ/RTI items are exactly where a missed date hurts most.
CPGRAMS: max redressal time 21 days (cut from 30 in Aug 2024); interim reply allowed if longer; appeals disposed within 30 days.	DARPG guidelines via pgportal.gov.in and press (retrieved). Verified.	Gives a defensible default SLA for CPGRAMS-sourced cases instead of an arbitrary number. AICTE internal SLA norms remain UNVERIFIED.
AICTE regulations: institutions must run online grievance portals; students may appeal to the Ombudsperson within 15 days of the committee decision; institutions report complaints in monthly status reports.	Institution compliance pages and Parliament replies (retrieved). Verified as stated by those pages.	Context for the problem; reporting data could be a future connector. Not built now.
AICTE has scheduled complaint-based surprise visits (SAC/SHC schedule marks some as COMPLAINT / SURPRISE VISIT).	AICTE circular, Feb-Mar 2019 (retrieved). Verified for that year.	Complaint history feeds inspection decisions. Justifies repeat-complaint institution view and Case Sheet.
AICTE bureau list beyond RB and PGRC; region list; Permanent ID format.	Not verified.	Remain labelled assumptions in seed data and slides. Say so openly.
Whether DARPG/CPGRAMS exposes an API AICTE could use.	Not verified.	Do not claim CPGRAMS integration. Claim a CSV/API ingest adapter that CPGRAMS exports could feed.

3. Final feature set (every item maps to the PS or to a verified officer pain)

A. PS core: all must be demonstrable

PS requirement	v2 feature
Centralised institution profile	Institution 360: Permanent ID, all cases across bureaus, bureau-wise counts, timeline, overlap banner

PS requirement	v2 feature
Case tracking fields	Case ID, receipt date, category, status, bureau, officer, last action, next action, last action date, expected disposal date on every case
Cross-bureau overlap detection, alert, linking	Tier A (open cases in 2+ bureaus) and Tier B (scored pairs, reason codes); link with lead bureau and note; banner on both cases; dismiss with suppression rule
Real-time dashboard	Clickable KPIs, bureau x status workload, trend, ageing buckets, beyond-SLA, open overlaps; 45 s polling
Workflow and action history	Append-only timeline: actor, bureau, time, remarks, transfers, escalations, attachments
Search and filters	Institution name, Permanent ID, case ID, bureau, region, category, status, date range
Alerts and notifications	Overlap, duplicate, reminder, due soon, overdue, escalation, hard deadline, unacknowledged transfer
Reports and analytics	Bureau, region, institution, ageing, disposal rate, status, overlaps, monthly, quarterly; CSV/JSON/PDF

B. Officer-workflow add-ons (each solves a concrete problem)

Add-on	Problem it solves	Honest limit
My Work Today	Officers open a dashboard and still do not know what to do first. Deterministic worklist: hard deadlines in 7 days, overdue, unacknowledged transfers, new overlaps, stale cases.	Only as good as the data fed in.
Hard-deadline tracking (CPGRAMS 21d, RTI 30d, court, PQ, VIP)	Court, Parliament and RTI dates are the costliest to miss. Separate from ordinary SLA.	RTI 30-day period is from the RTI Act (my knowledge, not re-fetched). Court/PQ dates are officer-entered.
Transfer acknowledgment	Transferred cases can sit unowned between bureaus. Receiving bureau must accept/reject; alert after 2 days.	2-day threshold is a placeholder.
Conflict Guard	Two bureaus issuing a show-cause notice or ordering inspection of the same institution. Warns before the action is saved; override needs a reason.	Rule-based on action tags; relies on officers tagging actions.
Same-bureau duplicate clustering	Mass complaints on one issue handled as separate cases.	Threshold tuned on synthetic data only.
Institution Case Sheet PDF	Parliament, court and review-meeting replies need a consolidated history fast.	Content is only as complete as ingested data.
Leadership view + weekly digest	Senior management needs region x bureau pressure points and repeat-complaint institutions without opening 50 cases.	Repeat thresholds are placeholders.

C. AI: three narrow uses, all optional and human-reviewed

- **Natural-language filter bar.** The LLM only converts a sentence into the same filter object the Cases page already uses; the UI shows the interpreted filters as chips. No generated SQL, no actions. Falls back to a keyword parser.
- **Institution brief.** 5-line summary built from structured fields with case numbers cited, labelled AI-generated.
- **Draft coordination note** when linking cases; editable, never auto-saved.
- **Privacy caveat to state honestly:** demo uses synthetic data. For real complaints, sending text to an external LLM is not acceptable without approval; the real deployment would need an on-premise model or redaction. The app works fully with AI switched off.

D. Deliberately NOT building

General chatbot (officers need answers from structured data, which filters and the worklist already give), complainant-facing app (outside this PS), voice, blockchain, WebSockets, automatic closing/transfer of cases, embeddings (future work), email/SMS (future work). Saying what you cut and why reads as judgment, not weakness.

4. How the app maps to the problem statement

Problem: the same institution can have complaints in RB, PGRC and other bureaus, with no shared view. Result: duplicated or conflicting action, delays, no accountability. **This app** pulls every case into one institution-centred view, flags when two bureaus are acting on the same institution, warns before conflicting actions, tracks deadlines (ordinary and hard), and keeps an append-only record of who did what. It sits above existing bureau systems through API and CSV intake and replaces none of them.

5. Workflow: how officials actually use it

Step	What happens	Who
1. Intake	Bureau system pushes cases via API key, or officer uploads CSV, or enters manually. Duplicate (bureau, case no) is ignored. Institution matched by Permanent ID, else fuzzy match; low-confidence matches go to a human review queue.	Bureau officer / system
2. Start of day	Officer opens My Work Today: hard deadlines, overdue, transfers waiting for acceptance, new overlaps involving their bureau.	All officers
3. Work a case	Open case, see all fields and timeline. Log action with a tag. If another bureau recently logged the same formal action on this institution, Conflict Guard warns first.	Bureau / regional officer
4. Overlap review	Overlap Center shows Tier A/B flags with score and plain reason codes. Officer compares two cases side by side, then links (choose lead bureau, add note) or dismisses. Banner appears on both cases.	Any involved bureau
5. Transfer	Case moved to another bureau; receiving bureau accepts or rejects with reason; unacknowledged transfers raise an alert.	Both bureaus
6. Monitoring	Worker raises due-soon, overdue, reminder, escalation and hard-deadline alerts; escalation writes an audit row.	System, senior mgmt
7. Reporting	Senior management opens Leadership view; any officer exports reports or an Institution Case Sheet PDF for a meeting, Parliament question or court reply.	Senior mgmt / all

6. Why officials benefit (claims you can defend)

- One place to see which bureau is handling which institution (no phone calls between bureaus to find out).
- Early warning before two bureaus act on the same institution.
- Hard deadlines cannot hide inside a general pending list.
- Every action has actor, time and reason, which supports accountability and audit.
- Meeting and Parliament-reply preparation reduced to one export. (Do NOT quote time saved; you have not measured it.)

7. Build phases (use ANTIGRAVITY_MASTER_PROMPT.md)

Phase	Deliverable	Owner (keep original split)	Cut priority
P0-P2	Repo, compose, schema, migrations, auth, RBAC	P4 repo/CI; P1 schema/auth	Never cut
P3-P5	Workflow with audit, cases/institutions APIs, intake, CSV, attachments	P1	Never cut (attachments can slip)
P6-P7	Synthetic data, resolver, Tier A/B overlap engine, evaluation script	P2	Never cut
P8-P9	SLA worker, hard-deadline alerts, dashboard/report/alert/worklist endpoints	P4	Never cut; leadership endpoints can slip
P10	Conflict Guard	P1 (hook) + P4	Cut after hard-deadline
P11-P14	Frontend: shell, My Work Today, cases, case detail, institution 360, dashboard, overlap center, alerts, reports, import	P3	Core never cut
P15	AI: NL filter, brief, draft note	P2 backend + P3 UI	FIRST to cut
P16-P17	Polish, Hindi labels, Playwright smoke, deploy, README, fresh-clone test	P3 + P4	Hindi labels cut before tests

Timeline assumption: I do not know your SIH deadline. If you have under 3 days, build P0-P9, P11-P14 and P16-P17 only; use Conflict Guard and AI as roadmap slides, not demo claims. Do not show a feature in the video that is not working.

Process rules that prevent the usual AI-coding failure: one owner per folder; one phase prompt at a time; run tests before commit; after each phase ask the agent to diff against AGENTS.md for invented fields; do not upgrade dependencies late; deploy a skeleton early. Hosting: backend needs a Docker host (Vercel alone will not run the API/worker/database); verify current free-tier limits before choosing.

8. Demo video (3 minutes) and PPT changes

Time	Show
0:00	Login as PGRC officer. My Work Today: a CPGRAMS case due in 2 days, a transfer waiting, a new overlap.
0:30	Open the overlap: score, plain reason codes, side-by-side. Link, choose lead bureau, add note. Banner on both cases.
1:15	As RB officer: log a show-cause notice on the same institution. Conflict Guard warning appears.
1:45	Institution 360 and Case Sheet PDF export. Dashboard KPI click-through to a filtered list.
2:15	Run clock-forward script: due-soon, overdue, escalation alerts appear; timeline shows system escalation.
2:40	Evaluation page: held-out precision/recall/F1 vs baseline, plus synthetic-data disclaimer. Close on the requirement checklist.

- PPT: add one slide 'Requirement-to-feature checklist' (Section 3A), one 'Officer workflow' (Section 5), and one 'What we deliberately did not build'. Put evaluation numbers ONLY from the script run.
- Every slide and the video must say synthetic data and placeholder SLA values. Avoid 'real-time' unless polling is working; say 'refreshes every 45 seconds'.

9. Risks and honest limits

- Synthetic evaluation numbers will look optimistic; state it. Have someone who did not write the matcher author the hard negatives and about 40 hand-written paraphrases.
- TF-IDF matching will miss cross-language complaints (e.g. Hindi vs English); say it is future work with multilingual embeddings.
- Bureau names, regions, SLA and thresholds are team assumptions, not AICTE norms.
- Nothing here guarantees selection.